Online Retail II – full report

This is the complete report behind the README: cleaning, methods, rules, checks, sensitivity tests and limitations.

The analysis focuses on two years of transactions (December 2009 – December 2011) from a UK-based online gift retailer that sells mostly to wholesalers. The data were cleaned in MySQL, studied in Python and presented in Power BI.

The project answers three business questions:

1. **When do customers buy?** → Cohort retention and seasonality.
2. **Who brings the revenue?** → Revenue concentration, customer segments and action lists for relevant accounts, tested on past data.
3. **Which products carry the business?** → Price bands, catalogue renewal and action lists for products.

Every choice in the analysis is justified by numbers in the notebook or the dashboard, tested for sensitivity where it matters, and its limits are stated.

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Key findings

- **Revenue is concentrated, and most key accounts stay.** The top 20% of products generated 77.8% of product revenue, the top 20% of customers 76.7% of known customers' revenue, and 60.5% of 2010's top customers were still in the top 20% in 2011.
- **Customers buy by the calendar, not by their age as customers.** Retention peaks every autumn whatever the month of the first purchase. The season comes in two waves, centred on 4 October and 18 November, and ends abruptly with the Christmas closure. November alone brings about half of the extra seasonal revenue (51.8% of September – December 2010; £798k in November 2011 against £771k in November 2010). The lesser October wave comes from many customers, not a few large accounts. Lines without customer ID behave like a separate channel and buy later, around 3 December.
- **Customers can be split into four segments** (*Core loyals*, *Lapsing regulars*, *Recent occasionals*, *Lost one-timers*). Core loyals are 22% of customers but 73% of customer revenue, and 81% of the top 20% customers; Lapsing regulars are another 17% of them.
- **264 customers, behind £1.27M of past revenue, are candidates for action**, in four lists with different owners: *Protect* and *Win back* for account managers, *Upgrade* for sales, *Retain* for automated campaigns.
- **The rules behind Protect and Win back passed tests on past data.** Regular customers flagged as overdue bought, in the next 6 months, only 31–43% of what they bought in the same months a year earlier (median), against 78–89% for the others. Customers judged probably lost by BG/NBD mostly were: at two dates with enough history, only 20–21% came back within 6 months (25 and 76 customers flagged), against 68–73% of the others. Of the 54 Protect customers, 43 bought in the same weeks a year earlier, so their silence is unusual; 2 were silent then too (maybe a seasonal pause), and for the shortest silences this check is weak.
- **Revenue held level while the existing range declined.** Product revenue did not change measurably in 2011: week by week, +2.0% ± 2.9%. But products still selling from 2010 lost 23.6% of their revenue over the same period, while those with no 2010 sales brought 20.8% of 2011 revenue. In pounds, year on year, products no longer sold lost £1,215k and products still selling £552k, while new ones added £1,949k, more than making up for both. These new products were launched in two waves (April – June and September – October) and were bought by the same key accounts as the rest of the range. The existing products lost buyers, not order size: each reached 26.7% fewer known customers, while those still buying ordered as much, at the same price. The most expensive price band was the weakest (−28.6% for its products still selling), and 46% of its 2010 products had no sales in the last three months.
- **Cheap products fill orders but bring little revenue.** The cheapest quarter of the catalogue moves 40% of the units, while only bringing 12.5% of the revenue. Customer segments do not differ in the price level of what they buy: they differ in how often and how much they order.
- **1,785 current products are candidates for action** (*Invest*, *Broaden*, *Refresh*, *Review*, *Investigate*, *Check customer*) from their growth and the breadth of their customer base; the priority lists contain the 264 of them in the top 20% of each price band.

Recommended actions

Account managers

- **Review the Protect list first** (54 Core loyals, £644k): these key accounts have missed two of their usual orders. The list is sorted by value, so the first calls are the most valuable.
- **Work the Win back list next** (43 Lapsing regulars, £445k): former regulars and buyers of large orders, who are probably lost.

Sales and marketing

- **Prepare for the season by early September.** The first wave peaks in early October: stock, catalogues and acquisition campaigns must be ready before. The channel without customer ID buys around early December, so keep the stock full until the Christmas closure.
- **Follow up the Upgrade list** (86 Recent occasionals close to the core): it is led by customers with large orders (up to about £4k per order), who are worth a personal contact while they are still active.
- **Automate the Retain list** (81 Recent occasionals, low value each): a reminder campaign is enough.
- **Target customers by how often and how much they buy, not by price.** Every segment buys about the same mix of cheap and expensive products: price-based offers would not separate them.

Product and category managers

- **Keep renewing the range.** Products with no 2010 sales brought 20.8% of 2011 revenue, while products still selling lost 23.6%. The high band needs it most.
- **Promote the Invest products** (96 in the priority lists, growing with a broad customer base) and compare their trend with similar products: a cheap test of whether the decline of the range is a life cycle or a lack of promotion.
- **Refresh the best sellers on the Refresh list** (111 in the priority lists): new variants or designs for products that still sell to a broad customer base, despite the decline.
- **Investigate the steepest falls** in the priority lists (18 Investigate and 5 Check customer products) before any decision: for products with a narrow base, first look at the orders of their main customer.

Data

- **Record the customer ID on every order**: it is missing on 22.8% of lines (15.7% of revenue), which are invisible to every customer analysis.
- **Tell returns apart from cancellations and record costs per product**: this would allow a quality analysis of returns and a ranking of products by profit.

Raw data

Online Retail II (UCI Machine Learning Repository): 1,067,371 transaction lines, 1 Dec 2009 – 9 Dec 2011, in two yearly sheets.

Columns: invoice, stock code, description, quantity, invoice date, unit price, customer ID, country.

Customer ID is missing on 243,007 lines (22.8%) and description on 4,382.

Cleaning (SQL)

The cleaning was done in MySQL 8, with every step logged in a `cleaning_log` table and re-runnable.

Step	Rows after	Rows removed	% of raw
0. Load and union of the two sheets	1,067,371	0	0.00
1. Overlap between the sheets	1,044,848	22,523	2.11
2. Text to uppercase	1,044,848	0	0.00
3. Exact duplicates	1,033,036	11,812	1.11
4. Price ≤ 0 and adjustment invoices ("A...")	1,027,016	6,020	0.56
5. Return lines	1,007,912	19,104	1.79
6. Purchases fully offset by returns	1,001,903	6,009	0.56
7. Final table: customer keys and product flag	1,001,903	0	0.00

- **Overlap.** The 22,523 sheet-1 lines in the time period also covered by sheet 2 were verified to be identical copies: no information was lost.
- **Duplicates.** Text was uppercased first, then the resulting exact duplicates were treated as recording errors and removed. Their weight is small: the extra copies were worth £57,481, at most 0.29% of final revenue, and only 38 customers (all small) lost more than 10% of their revenue because of this operation.
- **Prices and adjustments.** Lines with price ≤ 0 (free items, samples) were removed, together with the 6 manual adjustment lines (invoices "A..."), which held all 5 negative prices.
- **Returns.** Each return/cancellation was matched by a stored procedure to earlier purchases of the same customer, product and price, most recent first, and subtracted from them. Revenue was therefore removed at the purchase date, and cancelled orders did not inflate frequency or recency. 431,953 of 476,821 returned units (90.6%) were matched. The largest offsets were two huge cancelled orders (80,995 and 74,215 units). Thanks to this step, the seasonality and the growth of each product were measured on net sales at the right date, and the customer segments and action lists were not distorted by cancelled orders.
- **Missing customers** were kept as `MISSING-<invoice>` keys (22.8% of lines, 15.7% of revenue): they were included in revenue analyses, but excluded from customer analyses.
- **Non-product codes** (postage, fees, manual adjustments, tests) were flagged with `is_product = 0` instead of deleted (3.8% of revenue).
- **Missing descriptions.** No line without description was left after the cleaning steps above.
- **Products table.** A table was created with one description per stock code, the most frequent one (600 codes had more than one).
- **Exports.** `retail_export.csv` (the final table, without countries) and `products.csv` (the products table) were exported for the notebook in Python.

Final dataset: 1,001,903 lines, 39,750 invoices, 5,859 known customers, £19.73M revenue (£18.98M from products).

Validation

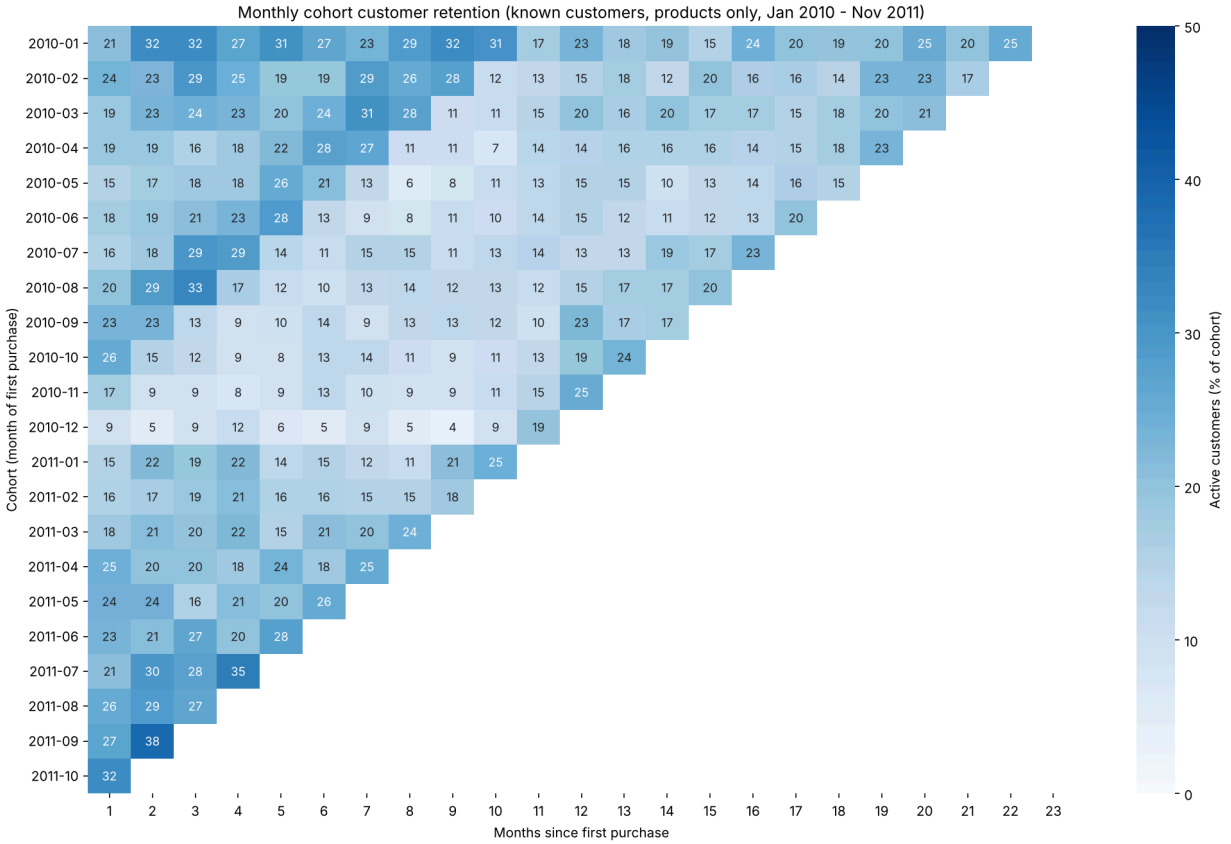
The export loaded in Python was checked in the notebook against the SQL results: every figure matches (1,001,903 rows, £19,727,380.03 revenue, 39,750 invoices, 5,859 known customers).

Analysis (Python)

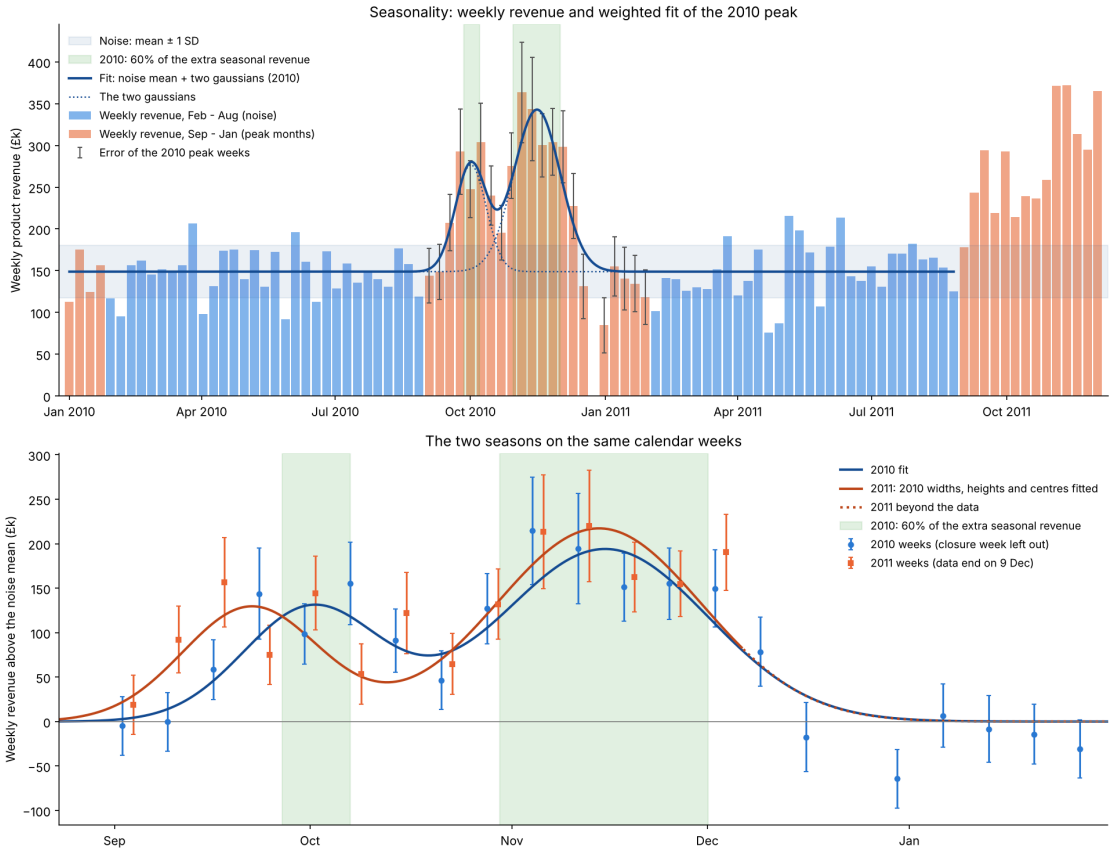
For the customer analyses only known customers and product lines were used (5,839 customers, £16.41M). Segment revenue shares refer to this total: Core loyals bring 73.2% of it, while they bring 63.2% of all product revenue, lines without customer ID included (the pie chart of the dashboard overview).

Years were counted from December to November, each containing a full Christmas season.

1. When customers buy: cohorts and seasonality



About 21% of new customers bought again the next month (known customers, product lines only) and about 18% were active 12 months later, but retention globally follows the calendar: the darkest diagonals of the heatmap fall in September–November for every cohort.



Weekly revenue was studied on 1 Jan 2010 – 31 Aug 2011, with the 2010 season near the centre. The peak months (Sep – Jan) were fixed beforehand from the cohort analysis, not from the fit, to avoid circular reasoning. Outside them (Feb – Aug) weekly revenue gave the baseline noise: mean £149k, SD £31k. Each peak week was corrected by subtracting the noise mean and was given an error combining the noise and the spread of its own purchases, $\sqrt{(\sigma_{noise}^2 + n \cdot s^2)}$. The 2010 peak was then fitted by weighted least squares, leaving out the Christmas closure week. Two gaussians fitted the peak well within its errors (reduced χ^2 0.92, $p = 0.54$, weighted R^2 0.83, correlation 0.93), one did worse (reduced χ^2 1.71, $p = 0.03$, weighted R^2 0.63, correlation 0.82). Therefore the 2010 season came in two waves, one around 4 October (30.5% of the extra revenue) and one around 18 November (69.5%). 60% of the extra seasonal revenue fell in two windows, 30 September – 10 October and 2 November – 4 December.

Extra revenue above the noise	2010	2011
September	£212k ± 79k	£377k ± 81k
October	£402k ± 78k	£407k ± 83k
November	£771k ± 106k	£798k ± 107k
December	£102k ± 80k	£254k ± 47k (1 – 9 Dec)

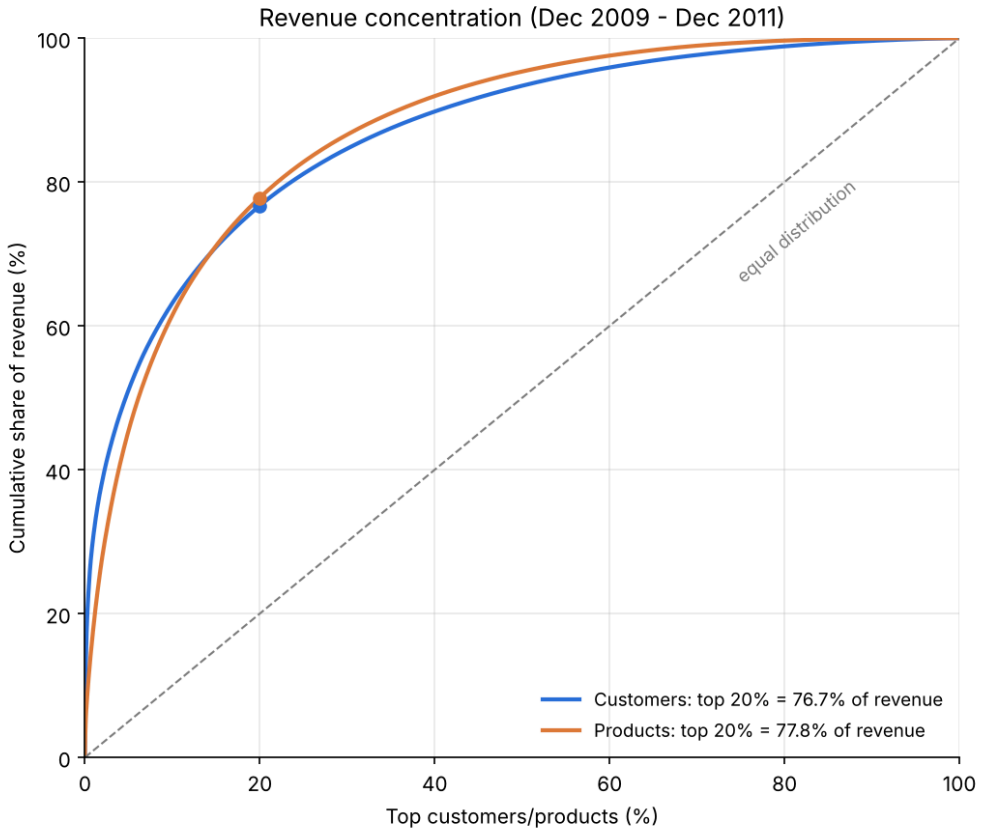
November alone gave half of the 2010 season (51.8% of September – December), and October and November were almost the same in both years.

The 2010 model also described 2011 (data until 9 December): with the 2010 shape and only the heights fitted, the reduced χ^2 was 1.54 ($p = 0.10$, not rejected at 5%) and the correlation 0.72.

Studying known customers (mostly wholesalers) and lines without customer ID separately, it emerged that the former followed the two waves of October and November, while the latter showed a single late wave around 3 December: their share of product revenue rose from 7.5–9.5% in September – October to 19–28% in November – December, in both years.

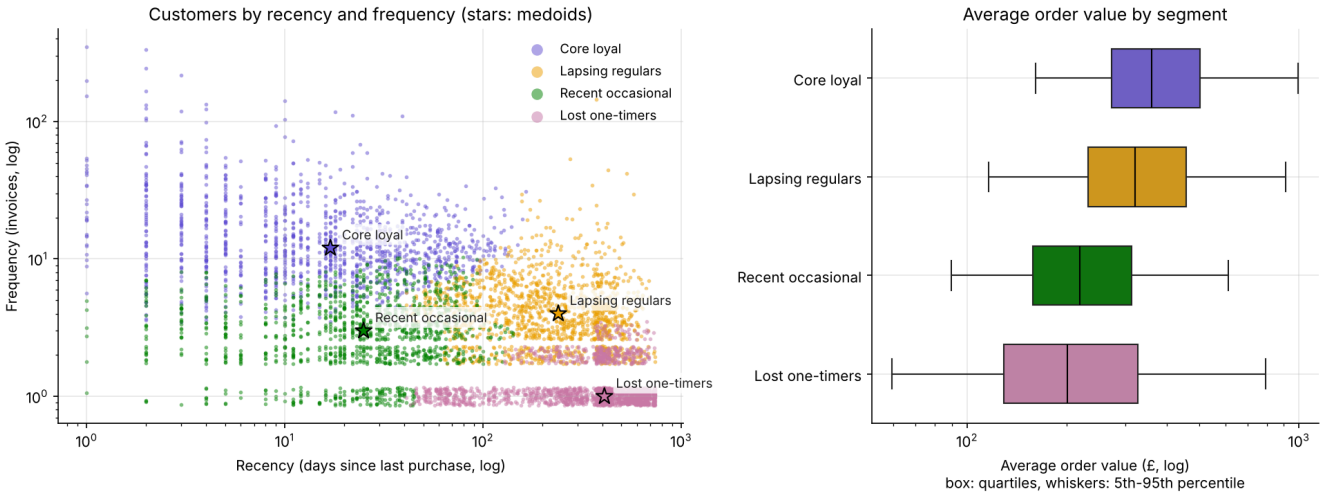
The next natural question to ask was whether the 2010 October wave was only the result of a few large wholesalers stocking up early, rather than a broad movement of many customers. Comparing, on known customers, an ordinary week (median of February – August) with the one full week inside the narrow first wave, it turned out that in the latter more customers bought (418 against 263), the 10 largest brought 18.4% of revenue against 29.0%, and without those 10 the week still made £190k, about twice the £93k of an ordinary week. The 4 weeks of the second wave looked the same (medians of 491 customers and 23.4%). Therefore both waves came from many more customers buying, not from a few large orders.

2. Revenue concentration and key-account stability



The top 20% of customers (1,168 of 5,839) brought 76.7% of known customers' product revenue. The result holds (76.7–77.7%) when customers without ID or non-product codes are included. Among 2010 top customers, 60.5% remained in the top 20% in 2011, and 2010's top customers as a whole generated 60.8% of known customers' 2011 product revenue.

3. Customer segmentation with my own k-medoids



Customers were described by recency, frequency and average order value, on product lines only (average order value replaced monetary, which is 0.85-correlated with frequency on the log scale). The features were log-transformed, standardised and compared with L1 distances. The clustering used my own k-medoids implementation (`kmedoids.py`, ported from a personal C project: k-means++ seeding, 10 restarts), and every resulting segment is represented by a real customer. The elbow and silhouette, computed on all customers and repeated with other seeds, cannot choose k (silhouettes 0.23–0.26 for k = 3–6). k = 4 is the smallest number of segments that separates the groups for the desired customer actions, based on recency and frequency, while k = 5 only further splits by order value. k = 4 was chosen.

Segment	Customers	Recency (days, median)	Orders (median)	AOV (£, median)	Revenue %
Core loyals	1,301 (22%)	17	12	361	73.2%
Lapsing regulars	1,476 (25%)	219	4	321	16.8%
Recent occasionals	1,254 (21%)	26	3	219	6.3%
Lost one-timers	1,808 (31%)	407	1	201	3.8%

The mean silhouette is 0.26: the segments are a useful partition for action, not natural customer types. Regarding stability, with other seeds the sum of the distances of the customers from their medoids barely changes (8,389–8,408 against 8,398 for the chosen solution), but only Core loyals are robust: 81–95% of them stay together, against 53–80% of each other segment. Some equally good solutions split inactive customers by order value instead of number of orders; the split by number of orders was kept because the Win back action looks for former regular customers.

Some information about the typical customer of each segment was extracted from the Power BI dashboard (customer detail page of the four medoids):

- **Core loyals (customer 14334)**. 12 orders from May 2010 to November 2011, one every month or two, each worth £224–558; last purchase 17 days before the end of the data.

- **Lapsing regulars (17142)**. 4 orders, in January, June and October 2010 and April 2011, then 240 days of silence. Their rhythm is slow, so this is only 1.62 of their usual gaps and they are very likely still active: the typical lapsing regular is not lost.
- **Recent occasionals (17777)**. A new customer, with 3 orders in August, October and November 2011.
- **Lost one-timers (14278)**. A single order of £205.83 in October 2010. BG/NBD gives P(alive) = 1 (the model treats them as active) to every customer without a repeat purchase: the model cannot flag one-timers, and the Win back action, based on BG/NBD, was therefore restricted to lapsing regulars.

4. Customer action lists, tested on past data

- Overdue measures recency in units of the customer's own average gap between orders. Two missed orders flag a customer (under a constant-rate purchase model, a longer silence has probability $e^{-2} \approx 13.5\%$): it is the early warning for Core loyals.
- P(alive), from the standard BG/NBD model (Fader, Hardie and Lee, 2005), is the probability that a customer is still active, given their number of purchases, the last one and the first one. Lapsing regulars below 40% are probably lost. P(alive) does not suit the other segments: Core loyals stay near 1 even after a long pause, and Recent occasionals have just bought.
- Silhouette towards another segment (below 0.1: close to that segment) shows which way Recent occasionals, whose few orders give no rhythm, are heading.

Action	Rule	Customers	Past revenue
Protect	Core loyal, overdue, in the top 20% customers	54	£643,806
Win back	Lapsing regular, P(alive) < 0.4, in the top 20% customers	43	£444,825
Upgrade	Recent occasional, close to Core loyals	86	£154,452
Retain	Recent occasional, close to Lost one-timers	81	£27,445

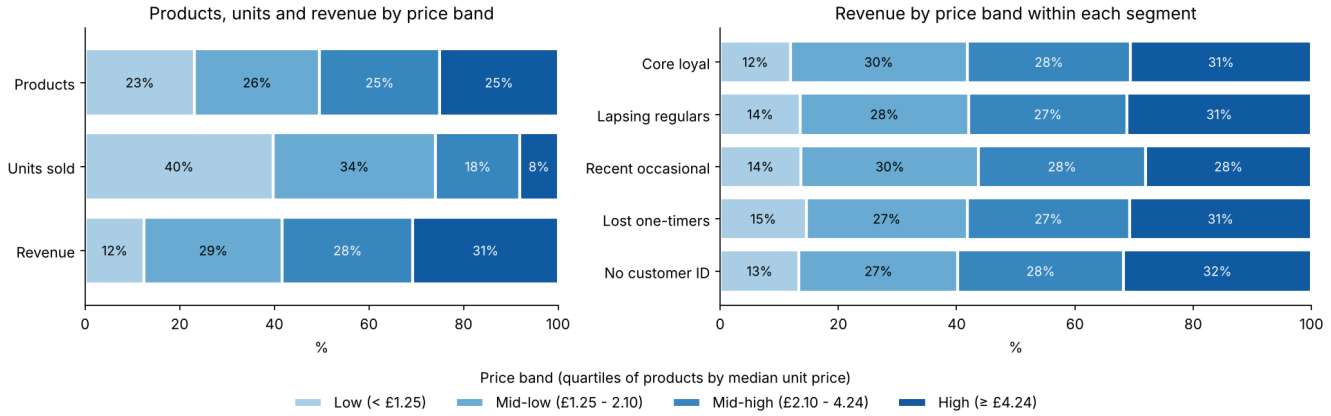
No Recent occasional is close to both Core loyals and Lost one-timers, so Upgrade and Retain never overlap.

Regarding the Protect action, the silence of an overdue Core loyal may simply be their usual pause right after the season. So each Protect customer's silence was moved back one year, to check whether they bought in that window: 43 of the 54 accounts (£553k of the £644k) did, so for them this silence is not their usual pause; 2 were silent then too; 9 were not customers yet. The check is weak for short silences: the shortest is 18 days (median 95.0 days), and a window of a few weeks a year earlier says little. The list was kept as it is, as a list of calls to check rather than of customers already lost.

As a backtest, both rules were applied, with the overdue and P(alive) thresholds fixed beforehand, at three past dates, from 1 Dec 2010 to 1 Jun 2011, with only the purchases before each date (BG/NBD was fitted again at each date; one small leak: returns were matched on the whole data, so a purchase before a date was already net of a return made after it). Each rule was checked on what it is meant to do. Overdue flags regular customers who are slowing down: among customers with 10+ invoices, those flagged placed, in the next 6 months, 31–43% of the invoices they placed in the same months a year earlier (median), against 78–89% for the others; comparing the same months of the two years was necessary to remove the effect of the season. P(alive) below 40% flags customers who are

probably lost: among customers with 3+ invoices, only 20.0% of those flagged in March 2011 (25 customers) and 21.1% of those flagged in June 2011 (76 customers) came back within 6 months, against 67.8% and 73.2% of the others. In December 2010 it flagged only 3 customers, too few to measure anything.

5. Price bands and catalogue renewal



Each product was given a reference price (median unit price paid by known customers: lines without ID pay about twice as much) and was put in a quartile band: below £1.25, £1.25–2.10, £2.10–4.24, £4.24 and up. All the sales of every band were then counted, whoever bought them. Pricing the bands on all lines would move 6.8% of the products, mostly those sold without ID, and leave the revenue by band almost unchanged.

Here are the lifecycle groups for the analysis of the catalogue renewal: *still selling* (sold in 2010 and in Sep – Nov 2011), *no recent sales* (sold in 2010, not in Sep – Nov 2011; removed, or just not ordered lately), *no 2010 sales* (new, or back after a pause). Growth was measured by comparing 2011 with 2010 on the same calendar period of each product, because the catalogue keeps changing: a product launched in mid-2010 has only a few months of 2010 sales and twelve of 2011, so comparing whole season years would show a much smaller fall than the real one.

Band	Growth of products still selling	2010 products with no recent sales	Products with no 2010 sales
Low	−21.1%	31%	117
Mid-low	−19.3%	37%	217
Mid-high	−23.9%	38%	168
High	−28.6%	46%	150
All	−23.6%	38%	652

Products with no 2010 sales brought 20.8% of 2011 revenue, but this percentage does not have the same base as the −23.6% of the products still selling, and the two do not add up. Whole year against whole year, the largest loss came from the products no longer sold:

Group	2010 (£k)	2011 (£k)	Change (£k)
Still selling	7,794	7,242	−552
No recent sales	1,385	170	−1,215
No 2010 sales	0	1,949	+1,949
All	9,179	9,361	+182

The revenue of the products still selling to known customers (−28.3% over the same period; the −23.6% above also includes the lines without ID) was then split into four factors, whose changes multiply:

Factor	Change, all bands	Share of the decline
Buyers (customer-product pairs)	−26.7%	93.1%
Lines per buyer	+0.1%	−0.2%
Units per line	−0.8%	2.4%
Revenue per unit	−1.6%	4.7%

It emerged that each existing product reached fewer customers, while those still buying it ordered as much, at the same price. This held in every band; only in the low band did the units per line also fall (−11.2%). The general decline therefore came not from discounts or smaller orders: customers stopped buying existing products, while products with no 2010 sales brought 20.8% of 2011 revenue. This fits a natural life cycle of gift items, with designs replaced by newer ones. Weak promotion of the existing range would also lose buyers, and these data cannot tell the two causes apart: promoting a few top existing products and comparing their trend with similar products would show which one it is (see the recommendations).

With equally healthy bands, 25% of each band's measurable products would be in the worst quarter of growth of all products (−54.6% or worse). They were 17.8% (Low), 21.8% (Mid-low), 30.0% (Mid-high) and 31.8% (High): the two most expensive bands declined the most.

Some information about the new products was extracted from the Power BI dashboard.

- The 652 products with no 2010 sales were launched in two main waves, 243 in April – June 2011 and 223 in September – October.
- Their monthly revenue grew from £6k in December 2010 to £477k in November 2011.
- They rested on the same key accounts as the remaining range: the top 20% customers of the Pareto analysis (over the two years) brought 76.3% of their 2011 revenue from known customers, against 76.8% of all 2011 product revenue from known customers.
- The channel without customer ID brought only 10.6% of their 2011 revenue, against 15.0% of all 2011 product revenue.

6. Product actions

Of the 4,722 products sold in the two years, 652 had no 2010 sales, as already said above, while 1,550 had no recent sales. Of the 2,520 remaining products (still selling), 1,785 were assigned one of six actions. These combine growth (compared with 0, with the catalogue's −23.6% and with the worst quarter of all products, −54.6%) and the breadth of the customer base (broad when no customer brings 25% or more of the revenue):

Action	Growth	Base	Products	Priority lists
Invest	growing	broad	435	96
Broaden	growing	narrow	136	19
Refresh	declining faster than the catalogue	broad	550	111
Review	declining faster than the catalogue	narrow	122	15
Investigate	in the worst quarter	broad	397	18
Check customer	in the worst quarter	narrow	145	5

Products declining less than the catalogue (383) and products too new to measure growth (352) got no action. Thresholds for actions were tested for sensitivity (broad-base share, months of history needed).

The table below shows how the products of each action are distributed across the four price bands.

Band	Invest	Broaden	Refresh	Review	Investigate	Check customer	All
Low	164	51	150	32	77	29	503
Mid-low	102	33	146	23	90	28	422
Mid-high	92	34	140	37	128	39	470
High	77	18	114	30	102	49	390
All	435	136	550	122	397	145	1,785

In revenue, the High band had the largest share in Investigate and Check customer (8.8% of its 2011 revenue, against 5.1–5.8% in the other bands), while the Low band had the largest shares in Invest (31.6%) and Refresh (27.5%).

Dashboard

The complete dashboard is in the file `powerbi/online_retail_dashboard.pbix`, built on `retail_export.csv` (the fact table) and the outputs below, with a date table in DAX and season years from December to November. Open it in Power BI Desktop (the data are included), or otherwise open the PDF export of every page (`powerbi/online_retail_dashboard.pdf`).

The overview counts product lines only: £18.98M of product revenue and the 5,839 customers who bought products. Its segment chart also includes the lines without customer ID and therefore counts Core loyals as 63.2% of product revenue, instead of 73.2% of customer revenue. The dashboard has six pages, each described below.

- **Overview**: product revenue, customers, invoices, the revenue share of the top 20% customers and products, and a note with the change of the products still selling on the same calendar period (−23.6%, from the notebook) and the meaning of the lifecycle groups; revenue by customer segment (with the lines without customer ID), revenue by month for each season year and revenue by product lifecycle; every chart filters the others.
- **Customers**: the segments (medians per customer), all customers by recency and frequency with the medoid of each segment, and the Pareto curve; a filter for the top 20% customers.
- **Customer actions**: the 264 customers in the four lists, with overdue and P(alive) highlighted where they cross the rule thresholds; a drill-through opens the monthly purchases of any customer.
- **Products**: price bands (units, revenue, products with no recent sales), the price mix of each segment, the lifecycle groups by band and the launches of the new products by month.
- **Product actions**: the actions by band, every current product by growth and dependence on its largest customer (with the rule thresholds), and the priority lists (in one big table); a drill-through opens the monthly revenue and the largest customer of any product.
- **Top 10**: the top customers and products of each season year, one year at a time.

The top 10 page shows that seven of the ten largest customers of 2010 were still in the top 10 in 2011, and that all of the 2011 top 10 are Core loyals. One of the three that left was customer 16754, ninth in 2010 with £52,900.99 and no purchase in 2011: they are in the Win back list. Customer 16029, the first of the Protect list, entered the top 10 in 2011. Six of the ten best-selling products stayed, among them 85123A, 22423 and 85099B. The top 10 customers brought 16.5% (2010) and 16.7% (2011) of known customers' product revenue, the top 10 products 7.9% and 8.2% of product revenue: the concentration of the Pareto analysis rests on a broad group of key accounts, not on a handful.

Outputs

- `output/customer_segments.csv`: all 5,839 customers with segment, action, RFM values and P(alive).
- `output/product_actions.csv`: the 1,785 products with an action, with description, band, priority flag and growth.

For Power BI, with `retail_export.csv` as the fact table:

- `customer_segments.csv` also holds overdue, the top 20% customers of the Pareto analysis and the medoid of each segment;
- `output/products_dim.csv`: every stock code of the sales lines with description, band, reference price, catalogue group, action, growth, share of the largest customer and Pareto flag.

What the actions mean

Here is what each action name means and what it asks for.

Customers (`customer_segments.csv`)

Action	Who	What it means	What to do	Owner
Protect	core loyal, overdue, in the top 20% customers (past revenue ≥ £2,850)	a key account has missed two of their usual orders	call first the most valuable ones (the list is sorted by revenue)	account managers
Win back	lapsing regular, probably lost, in the top 20% customers (past revenue ≥ £2,850)	a former regular among the most valuable customers has probably stopped buying	personal contact, with an offer if needed	account managers
Upgrade	recent occasional, close to core loyal	a recent, occasional buyer whose orders already look like a core customer's	personal follow-up while still active	sales
Retain	recent occasional, close to lost one-timers	a recent buyer whose orders look more like those of the lost one-timers	automated reminder campaign: low value each	marketing
No action	everyone else (5,575)	nothing specific to do now	—	—

Products (`product_actions.csv`)

Action	Growth	Base	What it means	What to do
Invest	> 0	broad	growing and sold to many customers	promote and keep in stock
Broaden	> 0	narrow	growing, but thanks to one large customer	offer it to similar customers to widen the base
Refresh	−54.6% to −23.6%	broad	still sold to many, but falling faster than the catalogue	renew it with new variants or designs; monitor the trend
Review	−54.6% to −23.6%	narrow	falling faster than the catalogue and dependent on few customers	check whether the fall is one customer's, then decide
Investigate	≤ −54.6%	broad	among the steepest falls, while many customers still buy it	find the cause (price, stock, design) before it drops out
Check customer	≤ −54.6%	narrow	a steep fall that may simply mean one customer bought less	check that customer's orders: keep it if they still need it, otherwise a candidate for exclusion

Limitations

Data

- Cancelled orders and real returns cannot be told apart, and many returns are partial: returns were not analysed as a signal of product quality.
- Customer ID is missing on 22.8% of lines: customer results cover 84% of revenue.
- "New" means new to these data (from 1 Dec 2009): long-standing customers look new in the first months (the December 2009 cohort was excluded), as do customers or products back after a pause; products with no recent sales may actually still be in the catalogue.
- December 2011 is incomplete (data end on 9 December) and was excluded from yearly comparisons; for the same reason the end of the 2011 season cannot be verified.
- Revenue is not profit: without costs, the Invest list only indicates the highest revenue potential, and all the lists are valued at past revenue, not at the revenue at stake.

Seasonality

- The two-wave shape is the best description of 2010, not a proven pattern, and two seasonal cycles are too few to validate a forecast honestly.

Customers

- Segments have weak separation (silhouette 0.26): only Core loyals are stable across seeds, and the borders of the other three segments are one choice among almost equally good partitions.
- Only the overdue and P(alive) rules were backtested, at three dates that are not independent tests. Segments were not rebuilt at past dates (clustering again at each date would do it), and whether a call or a campaign actually makes a flagged customer buy again cannot be measured with these data.
- Upgrade and Retain were not backtested, and their silhouette threshold matters most: 0.05 or 0.2 instead of 0.1 gives 40 or 233 Upgrade (28 or 167 Retain) customers.
- Since the lists were computed right after the season, some Core loyals who stocked up in November may look overdue without slowing down. BG/NBD, the standard model despite seasonal buying, assumes a constant buying rate; every Win back customer has been silent for at least 157 days, through the autumn peak, so the effect on them is small, but at earlier dates the model may flag customers who are only waiting for the autumn.

Products

- Product growth has noise of its own: a product just above 0% is "growing" for the rules, even if the change is not significant. The actions are prompts for a check, not decisions.

Further work

- Returns as a quality signal, if the data told real returns apart from cancelled orders: are the products in the worst quarter of declines also returned more often?
- A/B test of the action lists: contact half of the flagged customers and compare them with the other half, to measure what a contact changes.

- Margins: with costs per product, the product actions could rank by profit instead of revenue.
- Expected value of the customer lists: with a model of the order value on top of BG/NBD, the lists could be ranked by expected revenue instead of past revenue.
- A simple benchmark for the backtest: compare overdue and P(alive) with a plain rule, such as the days since the last purchase.

Repository structure

Folder or file	Content
`sql/`	`sql_cleaning.sql` : union of the two sheets, cleaning, checks and CSV exports
`python/`	`online_retail_notebook.ipynb` (the analysis, with its outputs), `py_utils.py` (helper functions), `kmedoids.py` (my k-medoids and silhouette functions)
`data/`	cleaned CSV files exported by SQL (not versioned)
`output/`	customer and product lists, and the tables for Power BI
`figures/`	charts produced by the notebook
`powerbi/`	Power BI dashboard (`online_retail_dashboard.pbix`) and its PDF version
`report/`	`REPORT.pdf` : this full report
`requirements.txt`	Python packages

To reproduce:

1. Download the original Excel file from UCI. In MySQL 8, create a database `online_retail` and load its two sheets into the tables `raw_2009_2010` and `raw_2010_2011` , with every column as text: `source_row` (the line number in the sheet), `invoice` , `stock_code` , `description` , `quantity` , `invoice_date` (`dd/mm/yyyy hh:mm`), `price` , `customer_id` , `country` .
2. Run `sql/sql_cleaning.sql` : it starts from the union of the two tables and exports `retail_export.csv` and `products.csv` . The export writes to MySQL's upload folder on Windows (`C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/`); on another system, change the two `INTO OUTFILE` paths to the folder of `secure_file_priv` .
3. Copy the CSV files to `data/` , install the packages with `pip install -r requirements.txt` and run `python/online_retail_notebook.ipynb` from top to bottom. It writes the lists to `output/` and the charts to `figures/` .